



Deep learning-based radiomics of computed tomography angiography to predict adverse events after initial endovascular repair for acute uncomplicated Stanford type B aortic dissection

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ABSTRACT

Purpose: This study aimed to construct a predictive model integrating deep learning-derived radiomic features from computed tomography angiography (CTA) and clinical biomarkers to forecast postoperative adverse events (AEs) in patients with acute uncomplicated Stanford type B aortic dissection (uTBAD) undergoing initial thoracic endovascular aortic repair (TEVAR).

Methods: We retrospectively evaluated 369 patients treated with TEVAR for acute uTBAD from January 2015 to December 2022. A three-dimensional (3D) deep convolutional neural network (CNN) automated radiomic feature extraction from CTA images. Feature selection, using Analysis of Variance (ANOVA) and the Least Absolute Shrinkage and Selection Operator (LASSO) algorithms, refined a radiomic score (Rad-Score). This score, alongside clinical parameters, was modelled via Extreme Gradient Boosting (XGBoost) analysis. Model calibration was assessed by calibration curves.

Results: The integration of the Rad-Score with clinical factors including albumin and C-reactive protein levels moderately enhanced predictive efficiency, exhibiting an area under the curve (AUC) of 1.000 (95%CI, 1.000–1.000) in the training cohort and 0.990 (95%CI, 0.966–1.000) in the internal validation cohort. In an independent validation cohort from another hospital, the combined model yielded an AUC of 0.985 (95%CI, 0.965–1.000), with an accuracy, precision, sensitivity, and specificity of 0.92, 0.92, 0.94, and 0.91, respectively.

Conclusions: The synergistic application of deep learning-based radiomics from CTA and clinical indicators holds promise for anticipating AEs post-initial thoracic endovascular aortic repair in patients with acute uTBAD. The clinical utility of the constructed combined model, offering prognostic foresight during follow-up, has been substantiated.

1. Introduction

Aortic dissection, typified by rapid onset, complex complications, and high mortality, is the leading aortic pathology [1]. The inflammatory process within the aortic wall is a recognized precursor to such conditions. Elevated biomarkers, including platelets (PLTs), C-reactive protein (CRP), and D-dimer are indicative of aortic dissection [2–4].

Computed tomography (CT), preferred for its accessibility, rapidity, and sensitivity remains the imaging modality of choice for suspected cases [5].

Type B aortic dissection (TBAD) manifests as either complex (cTBAD) or uncomplicated (uTBAD), with the latter commonly managed through thoracic endovascular aortic repair (TEVAR). Despite improved survival rates due to effective intervention, postoperative adverse events

Abbreviations: 3D, Three-dimensional; AE, Adverse event; AIC, Akaike information criterion; ALB, Albumin; ANOVA, Analysis of Variance; BMI, Body mass index; BNP, Brain natriuretic peptide; CNN, Convolutional neural network; CRP, C-reactive protein; CT, Computed tomography; CTA, Computed tomography angiography; cTBAD, Complicated type B aortic dissection; DCA, Decision curve analysis; FL, false lumen; GLCM, Grey level co-occurrence matrix; LASSO, Least Absolute Shrinkage and Selection Operator; PLTs, Platelets; ROC, Receiver operating characteristic; TBAD, Type B aortic dissection; TEVAR, Thoracic endovascular aortic repair; TL, true lumen; uTBAD, Uncomplicated type B aortic dissection; XGBoost, Extreme Gradient Boosting.

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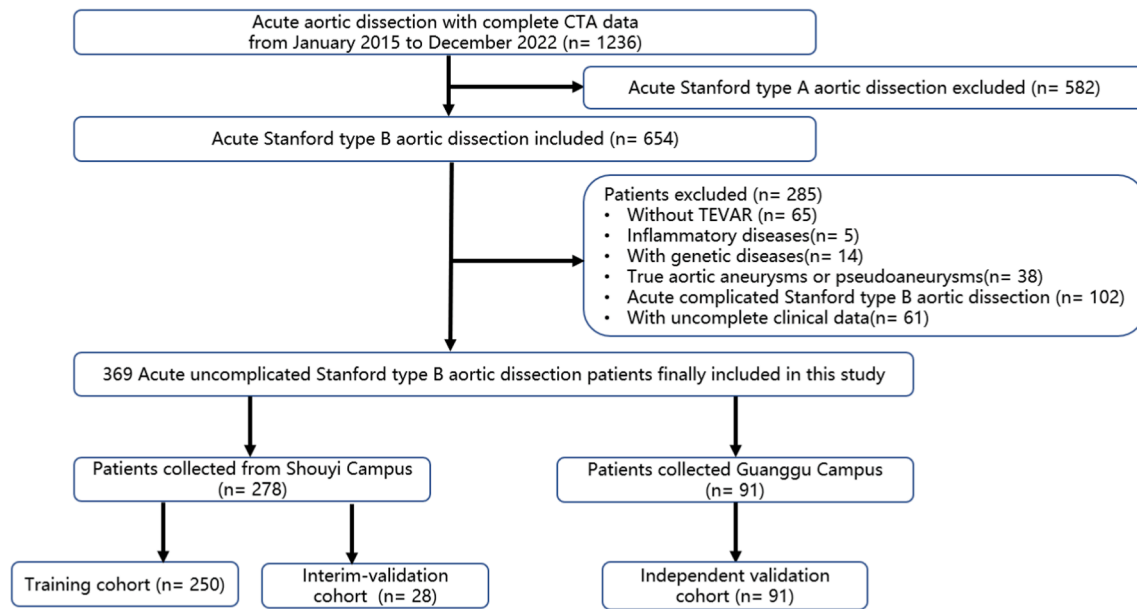


Fig. 1. Flowchart of uncomplicated Stanford type B aortic dissection patient selection in this study.

(AEs) can significantly diminish quality of life and increase mortality, underscoring the necessity for prompt AE detection and management [6].

In uTBAD, precise radiomic analysis of aortic CT data is critical. Contemporary research suggests that radiomic profiling can discriminate between different aortic dissections effectively [7]. However, the manual process is labour-intensive and subject to inter- and intra-observer variability [8]. Emerging deep learning techniques, especially U-Net-based three-dimensional (3D) convolutional neural network (CNN) architectures, are transforming the segmentation of aortic structures, offering substantial promise in monitoring and characterizing uTBAD [5].

This study proposes a novel predictive model that leverages deep learning-processed radiomic features from computed tomography angiography (CTA) in conjunction with clinical biomarkers to predict AEs following initial TEVAR in patients with acute uTBAD.

2. Materials and methods

2.1. Patients

This retrospective cohort study, conducted at Shouyi Campus of Renmin Hospital of Wuhan University, encompassed 654 individuals with acute uTBAD between January 2015 and December 2022. This study conformed with the tenets of the Declaration of Helsinki and was approved by the institutional review board of Renmin Hospital of Wuhan University Clinical Research Ethics Committee (Approval No. 2022 K-K083), and the requirement for informed consent was waived. Patient data were meticulously scrutinized to include those diagnosed with uTBAD via comprehensive aortic CTA prior to intervention, who received TEVAR, and who had intact clinical and laboratory data. The patients admitted to our facility underwent computed tomography angiography (CTA) within one week prior to surgery. This scheduling ensures that we possess the most recent anatomical data to guide the planning of the surgical procedure and allows for the evaluation of any acute alterations, thereby optimizing the surgical outcomes. Exclusions encompassed preoperative aortic inflammatory or genetic conditions such as Marfan syndrome, and prior true or pseudoaneurysmal aortic manifestations. AEs were defined in accordance with established literature criteria, encompassing aortic rupture, significant enlargement, aneurysm formation, ischemic complications, or emergent

hypertension/pain [6]. Ultimately, 369 patients were deemed eligible, with demographic details to be presented in subsequent sections.

For robust validation of the predictive model, we divided the cohort into training, interim-validation, and independent validation subsets. The 91 cases from Guanggu Campus of Renmin Hospital of Wuhan University constituted the independent validation group, while the remainder were randomly allocated to the training (250 patients) and interim-validation (28 patients) cohorts, using a 9:1 ratio (Fig. 1). There were no significant differences in gender distribution (Male/Female: 247/52 vs. 82/17, $p = 0.96$) or age (58.19 ± 12.45 vs. 60.03 ± 13.44 , $p = 0.27$) between the patient groups from the Shouyi Campus of Renmin Hospital of Wuhan University and the Guanggu Campus of Renmin Hospital of Wuhan University.

2.2. Imaging data

Aortic CTA was conducted using multidetector row scanners (64 to 256 slices) with varying specifications over the study period. This included the 64-slice Optima 680 and the 256-slice Revolution CT, both by GE Healthcare. A standardized protocol was employed, involving intravenous contrast with a flow rate of 3.5–4.0 ml/s and a total volume of 90 ml. Scanning covered the region from the thoracic inlet to the pubic symphysis, with parameters tailored to patient body mass index (BMI): 120 kV for BMI > 30 kg/m², 100 kV for BMI of 20–30 kg/m², and 80 kV for a BMI < 20 kg/m². The X-ray tube current was auto-adjusted, with a layer spacing of 5 mm and an acquisition thickness of 0.625 mm. Imaging window settings were 35–40 HU for the level and 300–350 HU for the width.

2.3. Deep learning modelling for aortic segmentation

Aortic CTA images were first extracted using an intensity range of 200–500 HU for contrast-enhanced arteries, and then resampled into volume of interest (VOI) with the same spacing and sample size [9]. We leveraged a 125-layer end-to-end 3D U-net architecture, augmented with DenseNet and IoU tracing as reported in Yu et al. [7] for the delineation of the entire aorta, true lumens (TL) and false lumens (FL), and the entry tear, with pre-processed VOI as inputs. This model, proven for its high precision achieving Dice similarity coefficients of 0.96 for TL and 0.95 for FL [9], utilizes a U-shaped 3D CNN for aorta segmentation, with IoU tracing delineating the TL and FL. Although the model

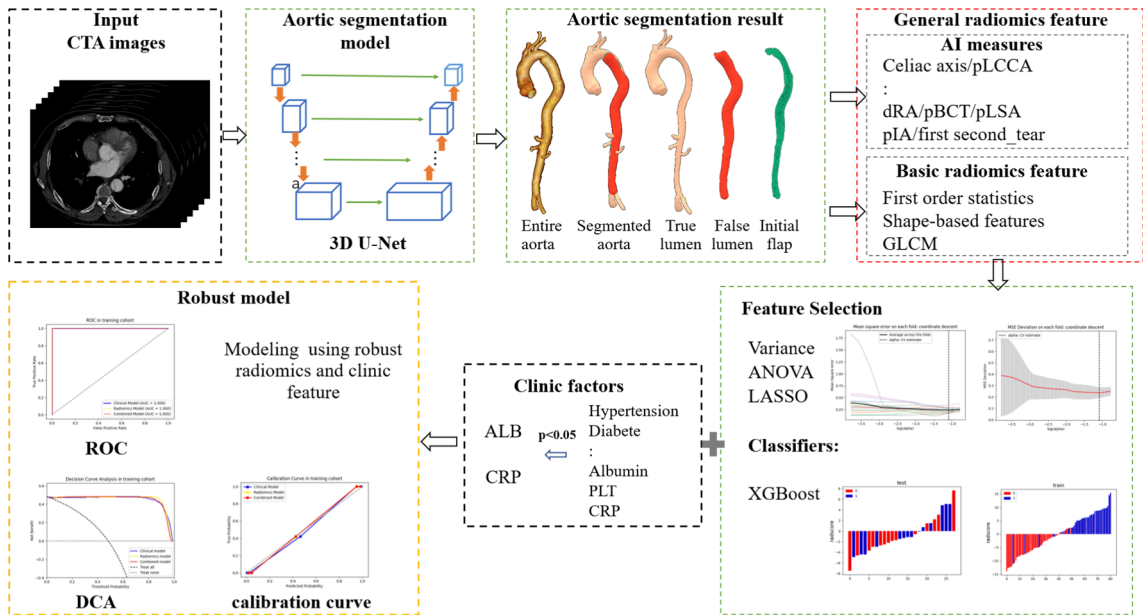


Fig. 2a. Overall adverse event-prediction modeling framework. **Abbreviations:** PLT, platelet; CRP, C-reactive protein; ALB, Albumin.

performed well in segmenting type B aortic dissection, it was less accurate in type A. The downsampling path features a $3 \times 3 \times 3 \times 32$ filter with a stride of 2, and each convolutional layer is succeeded by batch normalization and ReLU activation. The entire aortic lumen, which includes both the aortic wall and the blood volume occupied by the contrast medium were segmented and treated as region of interest for feature extraction.

Despite the high efficiency and precision of this model, discrepancies between automated segmentation results and actual anatomical structures were occasionally observed. When such discrepancies were identified, our medical team, comprising cardiovascular radiologists and vascular surgery experts with over five years of professional experience, manually adjusted the segmentation masks. These adjustments were guided by their expert knowledge in aortic anatomy and pathology. Specifically, the adjustments involved refining the boundaries of the TL and FL, ensuring that the segmentation accurately delineated the entry tear, and confirming that the segmented areas precisely matched the diseased sections of the aorta as visible in the CTA images.

The segmented lumens' diameters were calculated via a traversal

algorithm [8] (Fig. 2a). Our team standardized these measurements to ensure consistency and accuracy. Specific anatomical landmarks were identified for each measurement to minimize inter- and intra-observer variability. The actual parameters measured included the maximum and minimum diameters across various aortic segments and the volumetric assessments of the TL and FL, which are crucial for the calculation of radiomic features, including first-order shape and grey level co-occurrence matrix (GLCM) features for each luminal component. Our methodology ensures that these radiomic features accurately reflect the underlying aortic pathology, and therefore are reliable for inclusion in our predictive model.

2.4. Image-based feature extraction

Using the segmented aortic structures, we computed image-based features, including dimensions and radiomics [7]. The dimensions measured encompassed the maximum and minimum diameters across various aortic segments, and volumetric assessments of the TL and FL. Radiomics analysis yielded first-order, shape, and grey level co-

Table 1
Demographic and laboratory test results of enrolled patients.

Characteristic	Patients collected from one hospital			-	Patients collected from another hospital		
	No. (%)				No. (%)		
	Positive (n = 135)	Negative (n = 143)	P value		Positive (n = 47)	Negative (n = 44)	P value
Clinical information							
Age	58.82 ± 13.00	57.29 ± 12.22	.39		61.23 ± 14.41	59.11 ± 12.31	.36
Male	111 (82.22)	121 (84.62)	.59		39 (82.98)	37 (84.09)	.89
Hypertension	59 (43.70)	66 (46.15)	.68		19 (40.43)	14 (31.82)	.39
Diabetes	23 (17.04)	31 (21.68)	.32		12 (25.53)	9 (19.15)	.57
CAD	36 (26.67)	49 (34.27)	.17		11 (23.40)	16 (36.36)	.18
Laboratory test							
CRP	33.61 ± 26.34	24.18 ± 40.39	.01*		36.20 ± 20.41	23.68 ± 42.45	.02*
BNP	389.59 ± 364.00	363.81 ± 373.41	.77		311.18 ± 305.16	289.96 ± 290.45	.60
D-dimer	425.29 ± 2682.97	182.76 ± 244.69	.07		207.19 ± 376.73	158.01 ± 209.69	.05
PLT	219.02 ± 78.79	221.60 ± 84.88	.50		217.02 ± 71.89	217.55 ± 79.94	.43
NLR	8.76 ± 6.24	7.75 ± 4.28	.14		8.79 ± 17.69	8.79 ± 23.31	.66
WBC	52.20 ± 163.40	44.30 ± 170.47	.99		26.74 ± 12.09	17.86 ± 25.26	.24
ALB	38.93 ± 6.71	40.72 ± 5.12	.02*		34.69 ± 9.67	41.58 ± 6.06	.02*

***Abbreviations:** CAD, coronary artery disease; CRP, C-reactive protein; BNP, Brain Natriuretic Peptide; PLT, platelet; NLR, Neutrophil to Lymphocyte Ratio; WBC, white blood cell; ALB, Albumin.

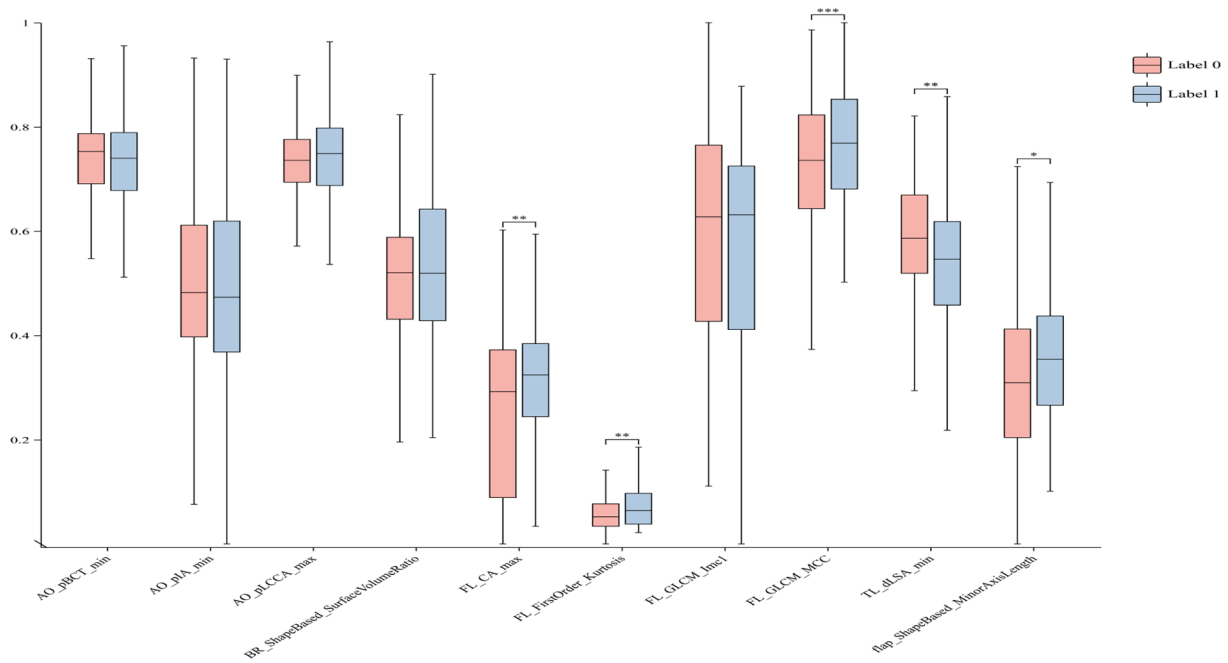


Fig. 2b. The boxplot of radiomics total features.

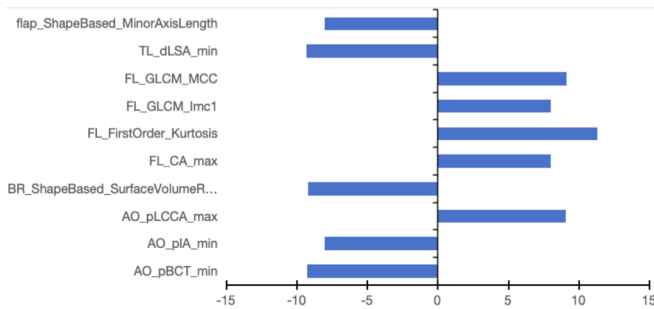


Fig. 2c. The fig of radiomics coefficient analysis.

occurrence matrix (GLCM) features for each luminal component.

The clinical factors encompassed demographic data, preoperative conditions, and laboratory results, such as complete blood counts and biochemical profiles. Relevant factors were identified via chi-square and Wilcoxon tests, and those with significance ($p < 0.05$) are highlighted in Table 1.

2.5. Feature selection and model validation

Dimensionality reduction of radiomic features was a three-step process, incorporating variance thresholds, Analysis of Variance (ANOVA) for statistical relevance, and Least Absolute Shrinkage and Selection Operator (LASSO) regression to pinpoint features with nonzero coefficients in AE prediction. Initially, the variance threshold value was set to 0.8, eliminating features with low variance. Afterwards, ANOVA was employed to identify radiomic features with significant differences between groups. Subsequently, the LASSO technique was applied to the features shortlisted by ANOVA to refine the selection further and mitigate overfitting by penalizing the absolute size of the regression coefficients. This process ensured the retention of the most relevant features. The resulting radiomics score (Rad-Score), representing a comprehensive quantification of the uTBAD characteristics, for each patient was calculated using LASSO-derived coefficients (Fig. 2b–c), assigning weights to each feature based on its contribution to the model's accuracy. Model validation occurred in both the interim-

validation and independent validation cohorts.

2.6. Interpretation

The integration of a deep learning model with clinical factors into a CTA-based combined model efficiently predicted post-TEVAR AEs. This approach can aid in patient stratification and individualized management of uTBAD, potentially enhancing therapeutic outcomes and minimizing unnecessary interventions.

2.7. Statistical analysis

Clinical factor comparisons between AE and non-AE groups were performed using appropriate univariate tests, followed by the Extreme Gradient Boosting (XGBoost) method for significant predictors. Model performance was appraised using receiver operating characteristic (ROC) curves, Akaike information criterion (AIC), calibration curves, and decision curve analysis (DCA). Statistical significance was recognized at $p < 0.05$. Analyses were executed on the R software (version 3.6.3).

3. Results

3.1. Demographic and laboratory test characteristics

All subjects in this study were followed up for 14.93 ± 16.90 months after TEVAR. AEs after surgery were as follows: fatal or non-fatal aortic rupture (6, 1.63 %), rapid aortic growth (>10 mm/year) (64, 17.34 %), aneurysm formation (≥ 6 cm) (62, 16.80 %), organ or limb ischemia (38, 10.30 %), new uncontrollable hypertension (13, 3.52 %), and new uncontrollable pain (20, 5.42 %). Some patients of this study had several of these manifestations at the same time. The demographic and laboratory test characteristics of the 369 patients enrolled in this study are summarized in Table 1, where positive and negative represent the occurrence and non-occurrence of AEs, respectively. Of the 369 included patients, 278 patients collected from one hospital were used as the training and interim-validation cohort, and the others collected from another hospital as the independent validation cohort. For the training and interim-validation cohort, compared with the negative group,

Table 2

The characteristics of the features.

Feature Name	Mean Value	Minimum Value	Maximum Value
CRP	28.76	0.50	200.00
ALB	39.85	21.49	57.90
AO_pBCT_min	33.22	0.00	45.03
AO_pIA_min	13.43	0.00	29.62
AO_pLCCA_max	33.69	0.00	45.28
BR_ShapeBased_SurfaceVolumeRatio	0.50	0.31	0.67
FL_CA_max	20.39	0.00	72.93
FL_FirstOrder_Kurtosis	8.73	0.00	98.14
FL_GLCM_Imc1	-0.15	-0.40	0.00
FL_GLCM_MCC	0.70	0.00	0.94
TL_dLSA_min	22.47	0.00	40.32
flap_ShapeBased_MinorAxisLength	54.65	0.00	162.74

***Abbreviations:** AO, aortic; ALB, Albumin; CA, celiac axis; CRP, C-reactive protein; dLSA, distal to left subclavian artery; FL, false lumen; GLCM, gray level co-occurrence matrix; pBCT, posterior to brachiocephalic trunk; Pia, posterior to iliac artery; pLCCA, posterior to left common carotid artery.

patients in the positive group had a higher C-reactive protein (CRP) (33.61 ± 26.34 vs 24.18 ± 40.39) and albumin (ALB) (38.93 ± 6.71 vs 40.72 ± 5.12), while no significant difference was found for all other clinical information between the two groups.

3.2. Adverse-event prediction model performance

3.2.1. Clinical model

ALB ($p < 0.05$) and CRP ($p < 0.05$) were significantly differently between the AE (+) group and AE (-) groups. There was no significant difference in platelets between the two cohorts.

3.2.2. Feature selection and radiomics model

Among the 261 first-order shape and GLCM image-based features from the training cohort dataset, seven features with low variance were deleted first, and 50 features remained after ANOVA. Then, the 10 most relevant features were finally selected using LASSO (Fig. 2). These features were incorporated into the construction of a radiomics signature, which includes the FL volume, the largest diameter of the FL in the sagittal plane and at the celiac axis (CA), the minimum diameter of the TL at the CA and at the position distal to the renal artery (dRA), and the sphericity of the initial flap (Table 2, Fig. 3). The models described in this study are available at: <https://bitbucket.org/hym-sk/model/src/main/>.

3.2.3. Combined model

The model was constructed via the XGBoost algorithm integrating the Rad-Score with clinical factors. There were significant differences in Rad-Score ($p = 0.001$), ALB ($p = 0.008$), and CRP ($p = 0.020$). A combined model was developed using the radiomics score with two clinical factors, namely ALB and CRP (Fig. 3).

3.3. Predictive performance of the radiomics, clinical, and combined models

The predictive performances of the radiomics, clinical, and combined models are summarized in Table 3. The ROC curves for each model are shown in Fig. 4. After combining the radiomic features and clinical factors, the area under the curve (AUC) values were higher in the independent validation cohort [0.985 (95 %CI, 0.965–1.000)] than that of the clinical model [0.954 (95 %CI, 0.916–0.992)] but slightly less than the radiomics model alone [0.987 (95 %CI, 0.973–1.000)]. The AUC of the combined model was significantly improved compared with

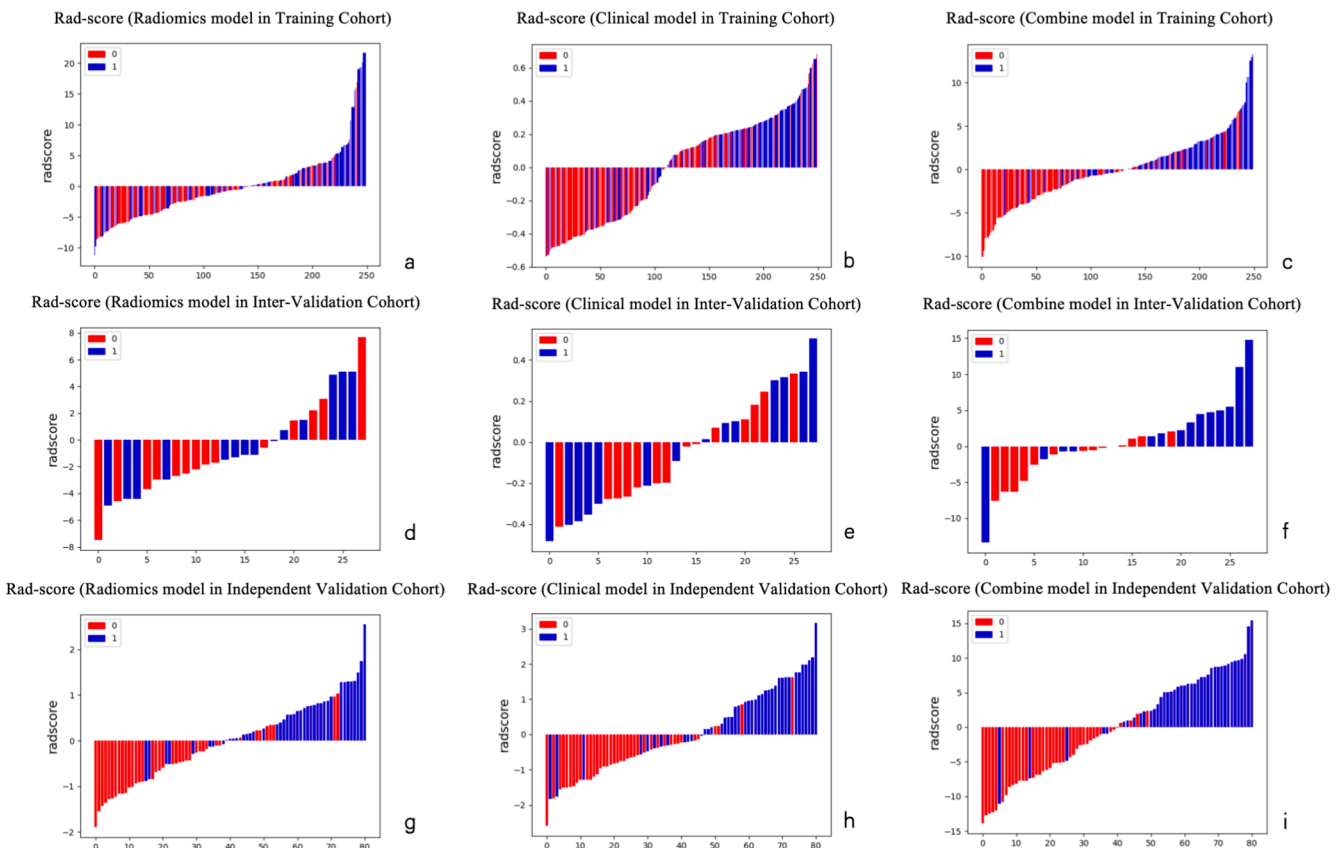


Fig. 3. Bar chart of risk score between different models. (a-c) radiomics model. (a) Training cohort. (b) Inter-Validation cohort. (c) Independent validation cohort. (d-f) clinical model. (d) Training cohort. (e) Inter-Validation cohort. (f) Independent validation cohort. (g-i) combined model. (g) Training cohort. (h) Inter-Validation cohort. (i) Independent validation cohort.

Table 3
The predictive performances of the radiomics, clinical, and combined models.

Metric	Model	Training cohort	Inter-validation cohort	Independent validation cohort
AUC	radiomics	1.000 (95 %CI, 1.000–1.000)	0.959 (95 %CI, 0.894–1.000)	0.987 (95 %CI, 0.973–1.000)
	clinical	1.000 (95 %CI, 1.000–1.000)	0.903 (95 %CI, 0.793–1.000)	0.954 (95 %CI, 0.916–0.992)
	combined	1.000 (95 %CI, 1.000–1.000)	0.990 (95 %CI, 0.966–1.000)	0.985 (95 %CI, 0.965–1.000)
Accuracy	radiomics	1.00	0.93	0.93
	clinical	1.00	0.82	0.85
	combined	1.00	0.93	0.92
Sensitivity	radiomics	0.99	0.86	0.87
	clinical	1.00	0.79	0.77
	combined	1.00	0.88	0.94
Specificity	radiomics	1.00	1.00	1.00
	clinical	0.99	0.86	0.93
	combined	1.00	0.93	0.91

*Abbreviations: CI, confidence interval.

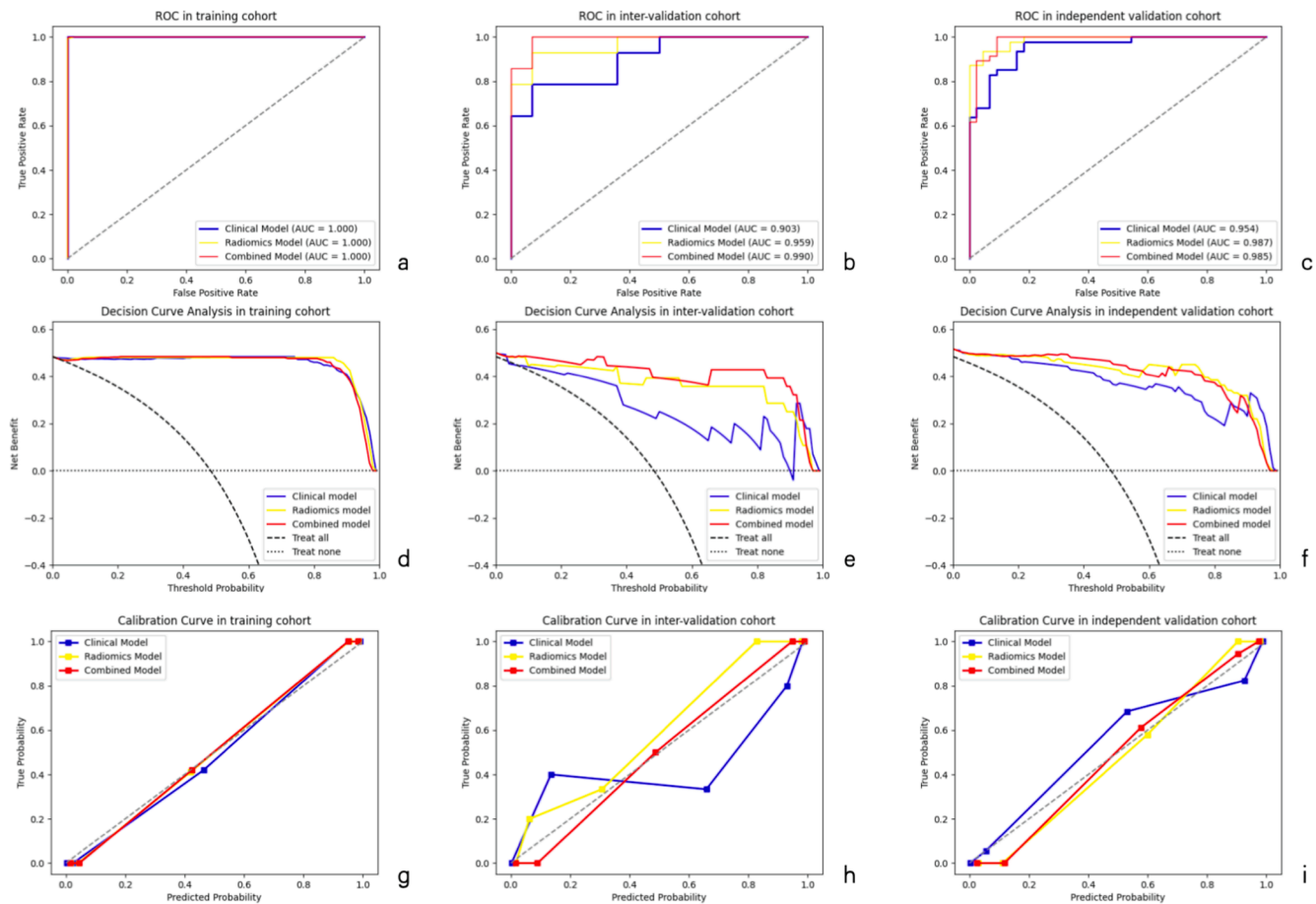


Fig. 4. Different adverse event – prediction models evaluation. (a–c) ROC of radiomics model, clinical model and combined model. The specific performance of combined model was shown AUC of 1.000 in training cohort (a), AUC of 0.990 in inter-validation cohort (b) and AUC of 0.985 in independent validation cohort (c). (d–f) DCA of radiomics model, clinical model and combined model (CI:95 %). The specific performance of combined model was shown in training cohort (d), inter-validation cohort (e) and independent validation cohort (f). (g–i) The calibration curve of radiomics model, clinical model and combined model in (g), inter-validation cohort (h) and independent validation cohort (i). Note: ROC, Receiver Operating Characteristic; AUC, area under the curve; DCA, Decision Curve Analysis.

the clinical model ($p = 0.04$), while there was no significant difference compared with the radiomics model ($p = 0.13$) and between the radiomics model and the clinical model ($p = 0.37$). Meanwhile, the sensitivity of the combined model [0.94] was greatly improved compared to either the radiomics [0.87] or the clinical [0.77] models. Moreover, the decision curve analysis (DCA) curves in the combined model also indicated its clinical usefulness, which is optimal compared to the other models (Fig. 4). The specific performance of the combined model demonstrated an accuracy of 0.92, precision of 0.92, sensitivity of 0.94, and specificity of 0.91.

4. Discussion

The management of acute uTBAD patients undergoing TEVAR grapples with the substantial risk of postoperative AEs [6,10], which can severely undermine the patient's quality of life. Our investigation delves into an integrated model that harnesses deep learning-based radiomics and clinical factors to provide a prognostic assessment of AEs with notable accuracy. This stratification into low and high-risk categories based on radiomic features is pivotal for the early identification of patients at risk, thereby enabling timely and effective clinical

interventions.

Our findings corroborate the critical role of morphological predictors such as the FL volume, its largest diameter on the sagittal plane, and the sphericity of the initial flap. These factors are indicative of a greater propensity for AEs, aligning with prior research that links increased FL size with escalated luminal pressure and potential rupture [6,11]. The intricate dynamics of blood flow near vital arterial branches like the celiac and renal arteries were also scrutinized, highlighting the predictive value of the TL's minimal diameter at these key junctures for AEs [6,12–20].

Clinical parameters, including ALB and CRP levels, were observed to herald postoperative AEs. These biomarkers, reflective of the underlying inflammatory, maintaining plasma osmotic pressure, and sustaining blood volume and prothrombotic state, could be harbingers of dissection progression or recurrence [21–24]. The activation and alteration speed of platelets can be relatively rapid, particularly in response to stimuli such as vascular injury or inflammation. Upon vascular injury, exposed tissues and substances can induce platelet adhesion and aggregation, leading to the formation of platelet thrombi aiming to repair the damage and achieve haemostasis [25–27]. Conversely, brain natriuretic peptide (BNP) levels necessitated conjunction with other risk indices to bolster predictive accuracy [28,29].

Although the radiomics model was highly specific on all cohorts and had a slightly higher AUC value and accuracy on the independent validation cohort, its sensitivity was less satisfied. The introduction of clinical parameters highly improved the sensitivity of the model, achieving our goal of early identification of patients at risk.

Despite our rigorous methodology, our study is not without limitations. For example the nature of the study was retrospective, and individual surgical variances could have influenced outcomes. Despite these limitations, our cohort represents a demographic consistently affected by uncomplicated aortic dissection and treated by a uniform surgical team. Prospective studies could enhance predictive precision by incorporating fluid dynamics and additional biomarkers into the model.

In summary, our study illustrates the potential of a deep learning-enhanced radiomics model, integrated with clinical indicators, to forecast AEs following initial TEVAR in acute aTBD patients. The established nomogram has proven its clinical utility and could significantly improve patient prognostication during postoperative follow-up.

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CRediT authorship contribution statement

Xuefang Lu: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Wei Gong:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Wenbing Yang:** Software, Resources, Methodology, Investigation. **Zhoufeng Peng:** Visualization, Validation, Data curation. **Chao Zheng:** Software, Resources, Formal analysis. **Yunfei Zha:** Writing – review & editing, Supervision, Conceptualization, Data curation, Funding acquisition, Methodology, Project administration, Resources, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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